Data Analytics Exercise 1: SQL Fundamentals

Query 1: Retrieve all columns from the Employees table. (Select Statement)

```
Select *
From 'Workspace.default.employees_table
Limit 10;
```

ID	First Name	Last Name	Department	Salary	HireDate	[redacted]	City
1	John	Doe	IT	55000	2018-06-15		New York
2	Jane	Smith	HR	48000	2019-07-20		Chicago
3	Mike	Johnson	Finance	60000	2017-09-30		Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25		New York
5	David	White	Marketing	52000	2016-04-10		San Francisco
6	Emily	Davis	IT	62000	2015-02-14		Chicago
7	Robert	Wilson	Finance	59000	2019-10-01		Houston
8	Jessica	Moore	HR	51000	2018-05-22		Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01		Chicago
10	Laura	Hall	IT	50000	2020-08-10		San Francisco

Query 2: Find all the unique departments in the employee table. (Distinct Statement)

```
Select Distinct department
From 'Workspace.default.employees_table
```

	Department
1.	IT
2.	HR
3.	Finance
4.	Marketing

Query 3: Retrieve all employees first & last names, ordered by salary in decending order. (OrderBy Statement): Select first_name, last_name, salary From 'Workspace.default.employees_table' OrderBy Salary Desc;

First Name	Last Name	Salary
Emily	Davis	62000
Mike	Johnson	60000
Robert	Wilson	59000
John	Doe	55000
Sarah	Brown	53000
Daniel	Clark	53000
David	White	52000
Jessica	Moore	51000
Laura	Hall	50000
Jane	Smith	48000

Query 4: Retrieve the Top 5 highest paid employees (Limit Statement)

```
Select first_name, last_name, salary
From 'Workspace.default.employees_table
Order By salary desc
Limit 5;
```

First Name	Last Name	Salary
Emily	Davis	62000
Mike	Johnson	60000
Robert	Wilson	59000
John	Doe	55000
Sarah	Brown	53000

Query 5: Employees who work in the IT department. (Where Statement)

First Name	Last Name	Department
John	Doe	IT
Sarah	Brown	IT
Emily	Davis	IT
Laura	Hall	IT

```
Select first_name, last_name, department
From 'Workspace.default'. Employees_table
Where department=IT;
```

Query 6: Employees who work in finance department Sal>58000 (AND Statement)

```
Select first_name, last_name, department, salary
From 'Workspace.default. Employees_table
Where department = Finance'
AND Salary>58000;
```

First Name	Last Name	Department	Salary
Mike	Johnson	Finance	60000
Robert	Wilson	Finance	59000

Query 7: Employees who work in the HR department Or Marketing Dept

```
Select first_name, last_name, department
From 'Workspace.default. Employees_table
Where department='HR' OR department='Marketing'
```

First Name	Last Name	Department
Jane	Smith	HR
David	White	Marketing
Jessica	Moore	HR
Daniel	Clark	Marketing

Query 8: Employees who do not work in the IT department (Where Not)

```
Select first_name, last_name, department
from 'Workspace.default.employees-table
Where Not department = 'IT';
```

First Name	Last Name	Department
David	White	Marketing
Robert	Wilson	Finance
Jessica	Moore	HR
Jane	Smith	HR
Mike	Johnson	Finance
Daniel	Clark	Marketing

Query 9: Employees which are in HR, IT, Finance departments (IN Statement)

```
Select * or first_name, last_name, department
from workspace.default.employees-table
Where department IN (HR, IT, Finance);
```

First Name	Last Name	Department
John	Doe	IT
Jane	Smith	HR
Mike	Johnson	Finance
Sarah	Brown	IT
Emily	Davis	IT
Robert	Wilson	Finance
Jessica	Moore	HR
Laurd	Hall	IT

(Combining Conditions)

1. Query 10: Employees who are in the IT department Sal > 50000 in NY

```
Select first_name, last_name, department, salary, city
from 'Workspace.default.employees_table
Where Department = 'IT'
AND Salary > 50000
AND City = 'New York'
```

First Name	Last Name	Department	Salary	City
John	Doe	IT	55000	New York
Sarah	Brown	IT	53000	New York

(Combining Where, AND, & OrderBy)

Query 11: Retrieve the first & last names of employees who work in finance or Marketing dep earn more than 52000 & order the results by salary in Desc order

```
Select first_name, last_name, department
from 'Workspace.default.employees_table
Where department IN ('Marketing', 'Finance')
AND Salary > 52000
Order By Salary Desc;
```

First Name	Last Name	Department	Salary
Mike	Johnson	Finance	60000
Robert	Wilson	Finance	59000
Daniel	Clark	Marketing	53000

Query 12: Find all unique cities where employees work, excluding IT, HR

```
Select Distinct
From workspace.default.Employees_table
Where department Not IN (IT, HR);
```

Department	City
Marketing	Chicago
Finance	Los Angeles
Marketing	San Francisco
Finance	Houston

Query 13: Retrieve employees who are not in the finance dept. have a salary greater than 50,000 and order the results by hire date in ascending order. (Combining Where, ~~OR, IN~~ NOT, AND and Order By

```
Select first-name, last-name, department, sal, hire-date
from 'Workspace.default.employees.table
Where Not Department='Finance'
AND Salary > 50000
Order By hire-date Asc;
```

First Name	Last Name	Department	Salary	Hire Date
Emily	Davis	IT	62000	2015-02-14
David	White	Marketing	52000	2016-04-10
Jessica	Moore	HR	57000	2018-05-22
John	Doe	IT	55000	2018-06-15
Sarah	Brown	IT	53000	2021-03-25
Daniel	Clark	Marketing	53000	2022-06-01

Query 14: First 3 employees who work in Chicago or Los Angele & belong In IT or Marketing

```
Select first_name, last_name, department, city
From 'Workspace.default.employees.table
Where City IN (Los Angeles, Chicago)
AND Department IN(IT, Marketing)
Limit 3;
```

First Name	Last Name	Department	City
Emily	Davis	IT	Chicago
Daniel	Clark	Marketing	Chicago